BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Riemann Integrability

Integrating Functions with Discontinuities

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CONDUCTED BY

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Continuous Functions are Riemann Integrable

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Let f be a continuous function on a closed interval $[a, b]$. Then f is Riemann integrable on $[a, b]$.

Monotone Functions are Riemann Integrable

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Let f be a monotone function on a closed interval $[a, b]$. Then f is Riemann integrable on $[a, b]$.

Discontinuous Function May Not Be Riemann Integrable

Dirichlet Function

Let f be the Dirichlet function defined on the interval $[0, 1]$ by

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Show that f is not Riemann integrable on $[0, 1]$.

Some Discontinuous Functions are Riemann Integrable

Discontinuity at one endpoint

Let f be the function defined on the interval $[0,1]$ by

$$f(x) = \begin{cases} 1, & \text{if } 0 < x < 1, \\ 0, & \text{if } x = 1. \end{cases}$$

Show that f is Riemann integrable on $[0,1]$ and compute its integral.

Discontinuity at a point

Let f be the function defined on the interval $[0, 2]$ by

$$f(x) = \begin{cases} 1, & \text{if } x \neq 1, \\ 0, & \text{if } x = 1. \end{cases}$$

Show that f is Riemann integrable on $[0, 2]$ and compute its integral.

Changing the value of a function at a point

Assume $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$. Show that if one value of f is changed at a single point $c \in [a, b]$, then the new function is also Riemann integrable on $[a, b]$ and the value of the integral does not change.

Functions with Finite Number of Discontinuities

Theorem

Let f be a bounded function on a closed interval $[a, b]$ that has only a finite number of discontinuities. Then f is Riemann integrable on $[a, b]$.

Changing value at finite number of points

Assume $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$. Show that if one value of f is changed at a finite number of points in $[a, b]$, then the new function is also Riemann integrable on $[a, b]$ and the value of the integral does not change.

Functions with Countable Number of Discontinuities

Theorem

Let f be a bounded function on a closed interval $[a, b]$ that has only a countable number of discontinuities. Then f is Riemann integrable on $[a, b]$.

Changing value at countable number of points

Assume $f : [a, b] \to \mathbb{R}$ is Riemann integrable on $[a, b]$. Show that if one value of f is changed at a countable number of points in $[a, b]$, then the new function may not be Riemann integrable on $[a, b]$.

Thank You!

We'd love your questions and feedback.

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References

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[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.